Sep 15, 2026 (Due: 08:00 Sep 22, 2026)

1. The 32-bit floating-point format discussed in the lecture is the IEEE-754 single precision format, which consists of 1 sign bit, 8 bits of exponent, and 23 bits of significand. Estimate the maximum and minimum (normal) finite floating-point numbers in this format.

Consider also the IEEE-754 double precision format, which consists of 1 sign bit, 11 bits of exponent, and 52 bits of significand. Estimate the maximum and minimum (normal) finite floating-point numbers.

2. Write a program to evaluate the harmonic series $\sum_{k=1}^{\infty} 1/k$ using IEEE-754 single precision floating-point arithmetic. What do you observe? Provide an explanation.

3. Let $\hat{x}$ be an approximation to x. In practice it is often much easier to estimate $\tilde{E}_{\text{rel}}(\hat{x}) = |x - \hat{x}|/|\hat{x}|$ compared to $E_{\text{rel}}(\hat{x}) = |x - \hat{x}|/|x|$. What is the relationship between E_{rel} and $\tilde{E}_{\text{rel}}$?

4. Suppose that you need to evaluate

$$f(x) = \mathrm{e}^{x} - \mathrm{e}^{-x} \quad \text{and} \quad g(x) = \frac{1}{\sqrt{1+x^2} - \sqrt{1-x^2}}$$

at $x \approx 0$. How to avoid cancellation?

5. You are given $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^n$, both already stored in floating-point format. Show that there exists a "small" matrix $E \in \mathbb{R}^{m\times n}$ such that $\mathrm{fl}(Ax) = (A+E)x$. Try to bound the entries of E as tightly as you can. You may assume that there is no overflow or (gradual) underflow in the calculation.

6. (H) Download OpenBLAS and LAPACK from the internet and compile them on your own computer. Run the test programs to ensure that you have successfully compiled the packages. You may write down the key steps and take some screenshots.

7. (H, optional) Kahan's summation algorithm is an accurate algorithm for computing $\sum_{k=1}^{n} x_k$. The algorithm reads

```
s = x[1]
c = 0
for k = 2 to n
  y = x[k] - c
  t = s + y
  c = (t - s) - y
  s = t
```

Construct a few examples involving cancellation to compare the accuracy of Kahan's summation algorithm with that of the naive summation method. Try to

provide a brief explanation of why Kahan's summation algorithm is more accurate.